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File Management

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`admin/verification/listUploadedFilesErrors.php``admin/verification/repairUploadedFiles.php``hclient/widgets/entity/manageRecUploadedFiles.js``hclient/widgets/entity/searchRecUploadedFiles.html``hclient/widgets/entity/searchRecUploadedFiles.js``import/biblio/syncZotero.php``import/biblio/zoteroMap.xml``import/fieldhelper/synchroniseWithFieldHelper.php``import/utilities/manageFilesUpload.php`

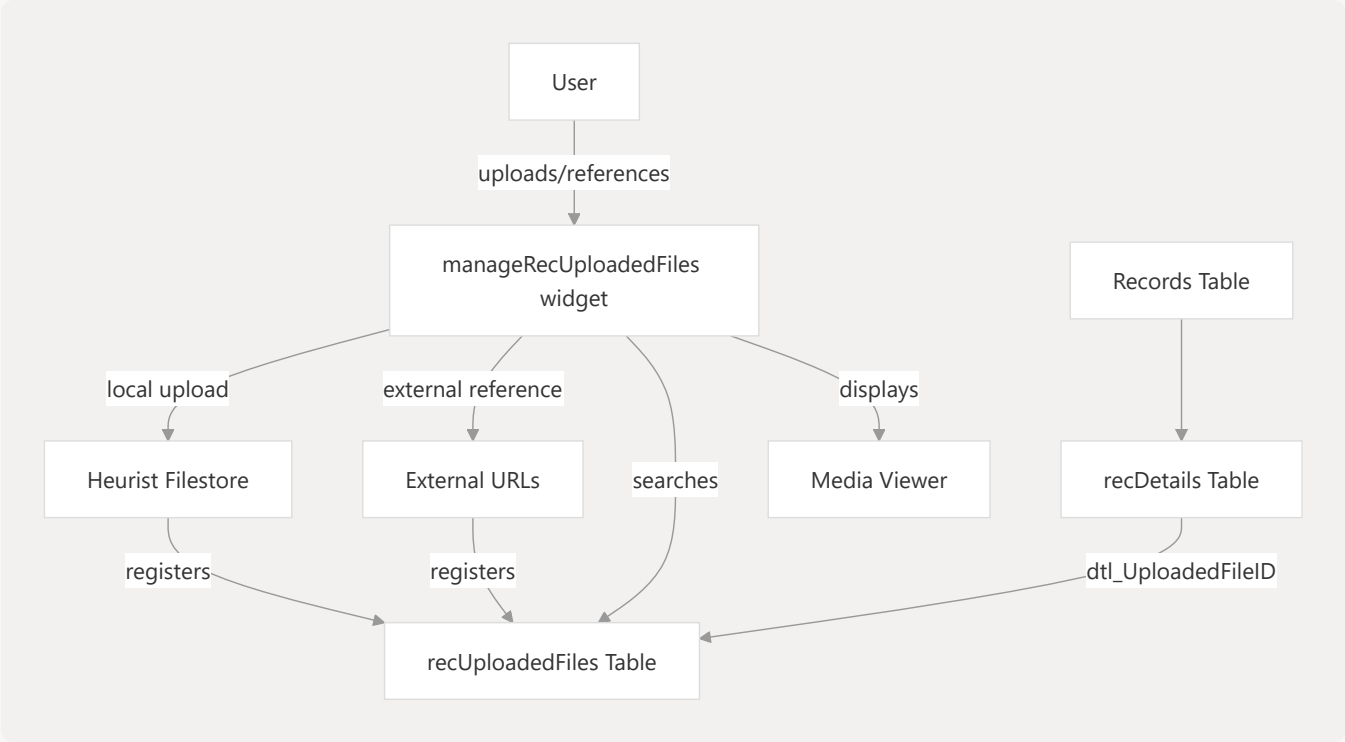
Purpose and Scope

The File Management system in Heurist handles the storage, retrieval, and management of files uploaded to or referenced by the system. It provides functionality for both locally stored files in Heurist's filestore and externally referenced files via URLs. This page documents the architecture, interfaces, and operations of the file management system. For information about importing data from structured files such as CSV or XML, see [Import System](#).

Architecture Overview

Heurist's file management system combines physical file storage with database records to track metadata and references.

File Storage Architecture



Sources: `hclient/widgets/entity/manageRecUploadedFiles.js` 21-304

`import/utilities/manageFilesUpload.php` 1-38

Database Schema

Files are represented in the database through the following tables:

Table	Purpose
recUploadedFiles	Stores metadata about files (local or external)
recDetails	Links files to records via <code>dtl_UploadedFileID</code> field
defFileExtToMimetype	Maps file extensions to MIME types

Key fields in `recUploadedFiles` :

Field	Description
<code>ulf_ID</code>	Primary key
<code>ulf_OrigFileName</code>	Original filename
<code>ulf_FilePath</code>	Path within filestore (local files)
<code>ulf_FileName</code>	Stored filename (local files)
<code>ulf_ExternalFileReference</code>	URL (external files)
<code>ulf_ObfuscatedFileID</code>	Unique ID for secure access

Field	Description
ulf_FileSizeKB	File size
ulf_Added	Upload timestamp
ulf_UploaderUGrpID	User who uploaded the file
ulf_MimeExt	File extension (external files)

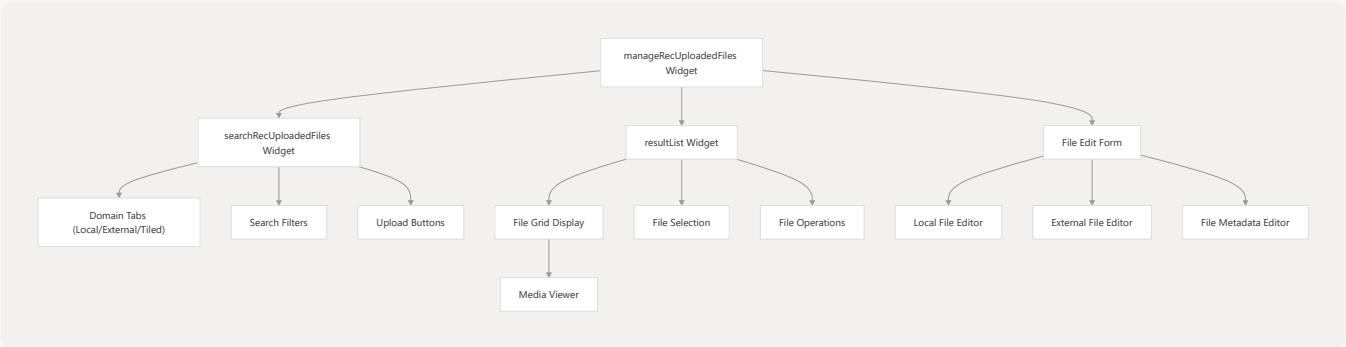
Sources: `hclient/widgets/entity/manageRecUploadedFiles.js` 210-302

`admin/verification/listUploadedFilesErrors.php` 332-370

File Management Interface

The interface for managing files is implemented through a set of widgets and components that allow users to upload, search, and manage files.

Interface Components



Sources: `hclient/widgets/entity/manageRecUploadedFiles.js` 45-154

`hclient/widgets/entity/searchRecUploadedFiles.js` 21-260

File Categories

Files are organized into three main categories accessed via tabs in the interface:

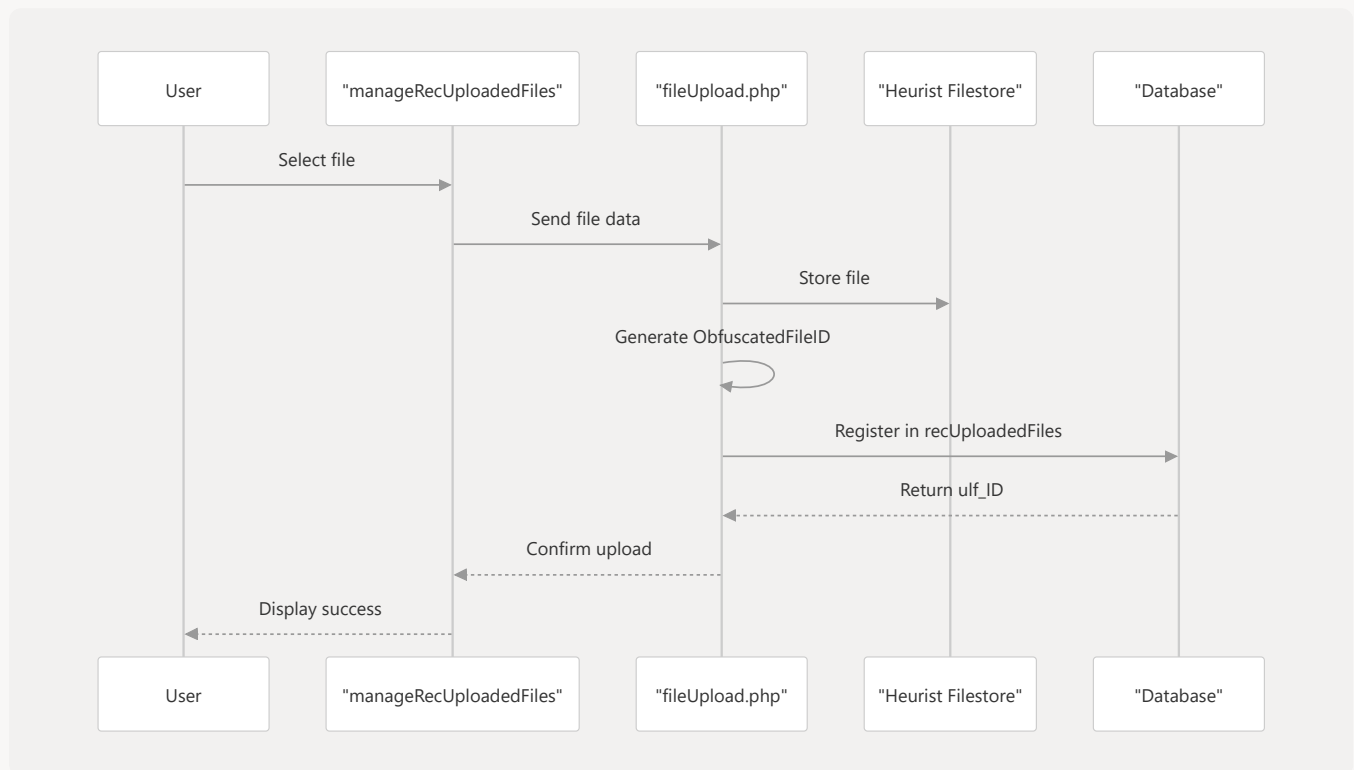
- 1. **Local Files:** Physically stored in Heurist's filestore
- 2. **External Files:** Referenced by URL
- 3. **Tiled Images:** Special handling for tiled image stacks (e.g., Zoomify, IIIF)

Sources: `hclient/widgets/entity/searchRecUploadedFiles.html` 92-99

`hclient/widgets/entity/searchRecUploadedFiles.js` 85-103

File Upload Process

Local File Upload Process



Sources: `hclient/widgets/entity/manageRecUploadedFiles.js` 434-444

`import/utilities/manageFilesUpload.php` 606-629

External File Reference Process

1. User enters a URL in the external file reference field
2. System attempts to detect the MIME type of the referenced file
3. File reference is registered in `recUploadedFiles` with `ulf_ExternalFileReference` field
4. No physical file is stored locally

Sources: `hclient/widgets/entity/manageRecUploadedFiles.js` 657-680

`hclient/widgets/entity/manageRecUploadedFiles.js` 192-207

Batch Upload

For bulk uploads, Heurist provides a specialized interface through `manageFilesUpload.php`:

1. Multiple files can be selected at once
2. Directory structure can be preserved when uploading
3. Files can be uploaded to specific folders configured in system settings
4. Duplicate handling options are available:

- Replace existing files
- Create new versions (adding numeric suffixes)
- Ignore changes to existing files

Sources: `import/utilities/manageFilesUpload.php` 234-320

`import/utilities/manageFilesUpload.php` 483-485

File Search and Filtering

The `searchRecUploadedFiles` widget provides comprehensive search capabilities:

Feature	Description
Text search	Find files by name or path
Type filter	Filter by MIME type (image, audio, video, etc.)
Reference filter	Show only referenced or unreferenced files
Sort options	Sort by name, size, or upload date
Owner filter	Show only files uploaded by current user
Domain filter	Switch between local, external, and tiled files

Sources: `hclient/widgets/entity/searchRecUploadedFiles.js` 110-139

`hclient/widgets/entity/searchRecUploadedFiles.js` 182-249

Media Creation

Heurist can automatically create Digital Media records for uploaded files using the `synchroniseWithFieldHelper.php` functionality:



The process maps file metadata to Heurist record fields:

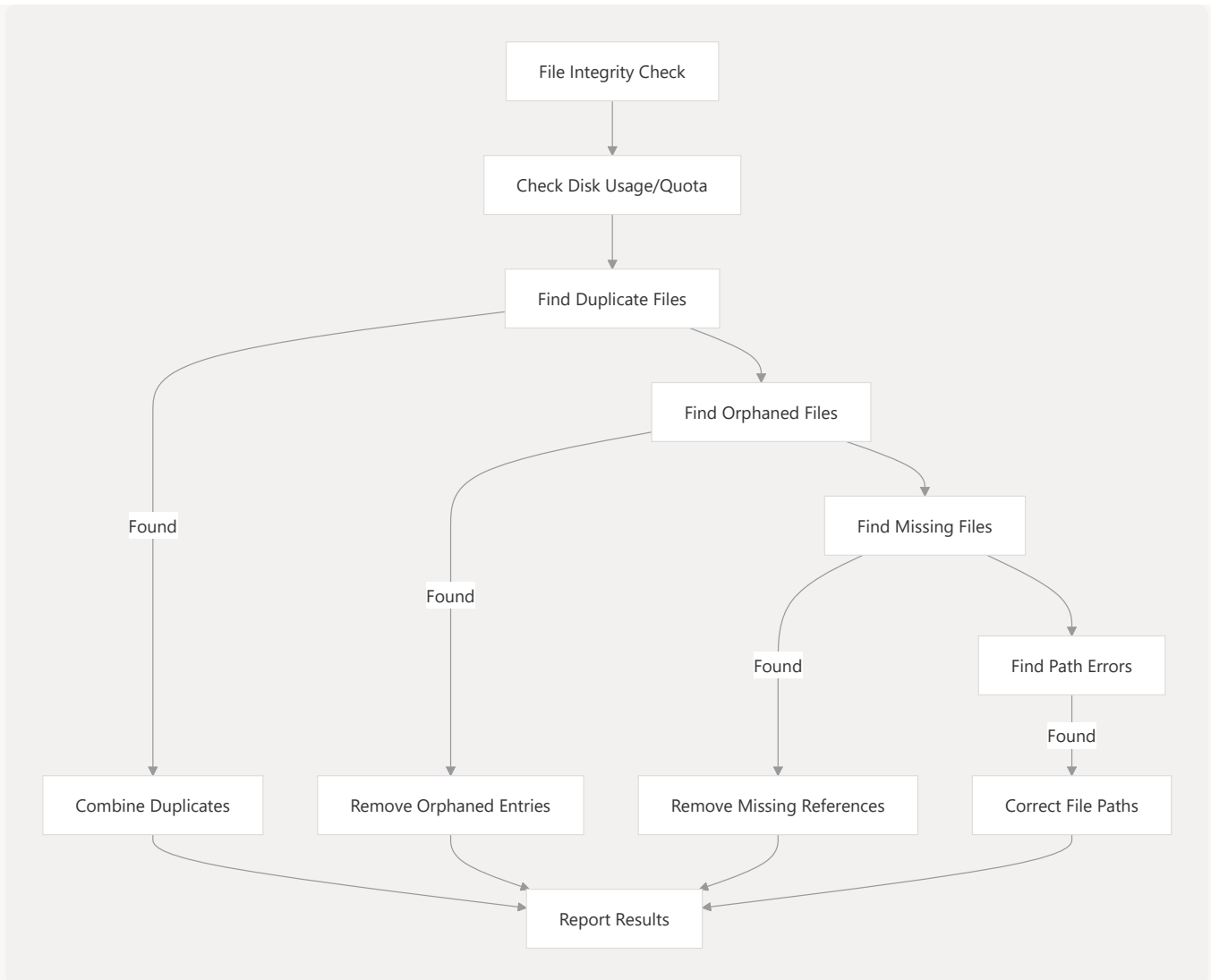
File Metadata	Heurist Field
Filename	Title (if no title available)
EXIF data	Various fields (date, description)
Geolocation	Geo coordinates
File path	File resource field
File size	File size field
MD5 hash	File hash field

Sources: `import/fieldhelper/synchroniseWithFieldHelper.php` 269-355

`import/fieldhelper/synchroniseWithFieldHelper.php` 400-608

File Integrity and Maintenance

Heurist provides tools to maintain file integrity through `listUploadedFilesErrors.php` and `repairUploadedFiles.php` :



The system checks for and can repair:

1. **Duplicate files:** Files registered multiple times, possibly with different paths
2. **Orphaned files:** Files in the database not referenced by any records
3. **Missing files:** Records referencing files that no longer exist physically
4. **Path errors:** Incorrect file paths that need to be corrected

Sources: `admin/verification/listUploadedFilesErrors.php` 126-214

`admin/verification/listUploadedFilesErrors.php` 330-478

`admin/verification/repairUploadedFiles.php` 68-135

Special File Types

Tiled Image Stacks

Heurist provides special handling for tiled image stacks:

1. Uploaded as zip files containing tiled images

- 2. Registered with special markers (`ulf_OrigFileName` starts with `_tiled`)
- 3. Displayed using specialized viewers that can navigate the tiled structure
- 4. Supports various tiling formats (Zoomify, IIIF, etc.)

Sources: `hclient/widgets/entity/manageRecUploadedFiles.js` 558-608

External Repositories

The system can connect to external repositories for file storage:

- 1. Files are uploaded to Heurist temporarily
- 2. Then transferred to the external repository
- 3. URLs are stored in `ulf_ExternalFileReference`
- 4. Accessed from the external repository when needed

Sources: `hclient/widgets/entity/manageRecUploadedFiles.js` 494-532

Implementation Details

The file management system is implemented across several components:

Client-Side Components

Component	Purpose
<code>manageRecUploadedFiles.js</code>	Main widget for file management
<code>searchRecUploadedFiles.js</code>	Search interface for files
<code>searchRecUploadedFiles.html</code>	HTML template for file search

Server-Side Components

Component	Purpose
<code>fileUpload.php</code>	Handles file uploads
<code>manageFilesUpload.php</code>	Interface for batch uploads
<code>synchroniseWithFieldHelper.php</code>	Creates media records
<code>listUploadedFilesErrors.php</code>	Lists file integrity issues
<code>repairUploadedFiles.php</code>	Repairs file records

